

$$A : \begin{matrix} \text{col}(A) \\ m \times n \end{matrix} \quad \text{col}(A^T) \quad [\text{row}(A)].$$

$$\text{null}(A) \quad \text{null}(A^T). \text{ [left null]}$$

$$\text{col}(A) \subseteq \mathbb{R}^m.$$

= set of all vectors that can be written as a lin comb. of cols of A .

$$= \left\{ \sum_{j=0}^n \underline{a}_j w_j : \{\underline{a}_j\} \text{ are the cols of } A, w_1, \dots, w_n \in \mathbb{R} \right\}.$$

$$= \left\{ A \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} : \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} \in \mathbb{R}^n \right\}.$$

Maximal lin indep set in $\text{col}(A)$ (i) Set of pivot cols in A .

Basis. (ii) Set of non-zero rows in $\text{rref}(A^T)$.

If there are k pivot cols in A : $\text{dimension}(\text{col}(A)) = k$.

$$\text{row}(A) = \text{col}(A^T) \subseteq \mathbb{R}^n$$

= set of all vectors that can be written as a lin comb. of rows of A .

$$= \left\{ \sum_{j=0}^n \underline{a}_j w_j : \{\underline{a}_j\} \text{ are the cols of } A^T, w_1, \dots, w_m \in \mathbb{R} \right\}.$$

$$= \left\{ A^T \begin{bmatrix} w_1 \\ \vdots \\ w_m \end{bmatrix} : \begin{bmatrix} w_1 \\ \vdots \\ w_m \end{bmatrix} \in \mathbb{R}^m \right\}.$$

Maximal lin indep set in $\text{col}(A)$ (i) Set of pivot cols in A^T .

Basis. (ii) Set of non-zero rows in $\text{rref}(A)$.

If there are k pivot cols in A : $\text{dimension}(\text{col}(A)) = k$.

$$\text{dimension}(\text{col}(A^T)) = k.$$

If there are k pivot cols in A .

$$\begin{matrix} A \\ m \times n \end{matrix} = \begin{matrix} P \\ m \times k \end{matrix} \begin{matrix} C \\ k \times n \end{matrix}.$$

(P : pivot cols. C : matrix of coeffs)

every row of A = linear comb of rows of $C \Rightarrow \text{dim}(\text{row}(A)) \leq \text{dim}(\text{col}(A))$

A has k pivots. ($n-k$ free cols).

$$\text{null}(A) \subseteq \mathbb{R}^n$$

= Set of all solutions for $A\underline{z} = \underline{0}$.

$$= \{ \underline{z} : A\underline{z} = \underline{0} \}.$$

dimension of the null space. $\geq n-k$. (from what we said so far in class).

But in fact $\dim = n-k$.

Orthogonal complements.

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